APPLIED MULTIVARIATE METHODS
833:525 / 970:527

Umer Farooq
Homework 04 (50 points)

- *HW must be **typed** in an appropriate word processing software application.*
- *Responses should be numbered and in chronological order.*
- ***Indicate your full name on the top of the document.***
- ***Save a PDF of the completed document** with your first initial and last name, i.e., HW04_swilliams.pdf*
- ***Upload** the HW via Canvas before the due date.*
- ***Show only key steps of your calculation.** You get partial credit for showing work in case of wrong answer.*
- *Any hypothesis test must include a null and alternative hypothesis, test statistics, degrees of freedom (if applicable), decision and interpretation.*

Problem 1:

You were asked to help with analysis of birth weights (BW) of 10,000 infants born in Philadelphia during a certain period of time. The aim of the analysis is to see whether the birth weights of the infants are associated with mother's age at birth (continuous variable in years), mother's weight (continuous variable in kg), mother's smoking status (the maternal smoking status contains 4 categories "Non-smoker", "Past-smoker", "Passive-smoker", "Smoker"), and mother's home region, consisting of the 6 most populated areas in Philadelphia ("Center City", "South Philadelphia", "Southwest Philadelphia", "West Philadelphia", "Lower North Philadelphia", and "Upper North Philadelphia").

a) How many dummy variables do you need to create to analyze the effect of maternal smoking status? Select "Non-smokers" as a reference category. Write out all the dummy variables. (2 points)

   Since there are 4 categories in the maternal smoking status ("Non-smoker", "Past-smoker", "Passive-smoker", "Smoker"), and we are taking "Non-smoker" as the reference category, we need to create 3 dummy variables to represent the other three categories.
   PastSmoker: 1 if the mother is a past-smoker, 0 otherwise.
   PassiveSmoker: 1 if the mother is a passive-smoker, 0 otherwise.
   Smoker: 1 if the mother is a smoker, 0 otherwise.

b) **[Pick one correct answer]** How many dummy variables do you need to create to analyze the effect of region with Center City selected as the reference category? (2 points)
   i. 3
   ii. 4
   iii. Ans. 5
   iv. 6
   Since there are 6 regions in city. By k-1 formula we need 5 dummy variables.

c) Write down the population model that estimates birthweight based on variables age, weight, maternal smoking status and region. Make sure that it is clear what each predictor means. (6 points)

$$BW=\beta_0+\beta_1\times Age+\beta_2\times Weight+\beta_3\times PastSmoker(dummy)+\beta_4\times PassiveSmoker(dummy)+\beta_5\times Smoker(dummy)+\beta_6\times SouthPhiladelphia(dummy)+\beta_7\times SouthwestPhiladelphia(dummy)+\beta_8\times WestPhiladelphia(dummy)+\beta_9\times LowerNorthPhiladelphia(dummy)+\beta_{10}$$

$\times UpperNorthPhiladelphia(dummy)$

Each Predictor:
BW: Birth Weight of the infant.
Age: Mother's age at birth (years).
Weight: Mother's weight (kg).
Past Smoker: Dummy variable for mothers who are past-smokers (1 = Past-smoker, 0 = otherwise).
Passive Smoker: Dummy variable for mothers who are passive-smokers (1 = Passive-smoker, 0 = otherwise).
Smoker: Dummy variable for mothers who are smokers (1 = Smoker, 0 = otherwise).
South Philadelphia, Southwest Philadelphia, West Philadelphia, Lower North Philadelphia, Upper North Philadelphia: Dummy variables for mother's home region (1 = belongs to the region, 0 = otherwise). "Center City" is the reference category. It means (1 = belongs to the South Philadelphia, 0 = otherwise) (1 = belongs to the Southwest Philadelphia, 0 = otherwise) (1 = belongs to the West Philadelphia, 0 = otherwise) (1 = belongs to the Lower North Philadelphia, 0 = otherwise) (1 = belongs to the Upper North Philadelphia, 0 = otherwise)

β_0 is the intercept means birth weight of children without taking age and weight of mother at time of birth without even considering smoking status and region they are born in (which means without any variable), β_1 is the impact of mother's age on the birthweight of children $\times Age + \beta_2$ is the impact of Mother's weight on Child's weight at birth$\times Weight$ β_3, β_4 β_5 *takes the impact of past smoking, passive smoking and present smoking on children's birthweight.* $\mathcal{B}_6$ β_7 β_8 β_9 β_{10} takes the weight of children at birth born in different regions of Philli.

Problem 2:

Dataset to be used: `eblcat.csv`

Please do not hand in computer output separately from responses to the questions below. Instead, cut and paste relevant output. Copy and paste (only relevant) R codes at the end of your homework in an appendix (annotated and in chronological order).

eblcat.csv is a dataset containing demographic, preoperative, and intraoperative information for 142 pancreatic resections for pancreatic cancer. It was designed to evaluate relationships between preoperative and intraoperative factors and blood loss during surgery.

Below is a brief description of the variables you will use from the eblcat.csv dataset:

ebl = estimated intraoperative blood loss, expressed in mL
racecat = race/ethnicity (0=White 1=Black 2=Asian 3=Hispanic)
les_size = tumor size in cm

Note: this data is for the purpose of illustration.

a) We are interested in whether race significantly changes the effect of tumor size on blood loss. How will you go about assessing this analytically? You may respond in a bulleted list. (5 points)
   - First Generating dummy variables for Race by keeping White as 0, Black as 1, Asian as 2 and Hispanic as 3. Means if race Black 1 otherwise 0 if race Asian 1 otherwise 0 if race hispanic 1 otherwise 0. By k- it will be 3 dummy variables
   - Later, estimating a model keeping an interaction term of race and tumor size because question asked if race significantly effect tumor size. We use both race dummies as well as their interactions for our analysis.
   -
```
setwd("C:/Users/HP PAVILLION 17/OneDrive/Desktop/Mult")
library(readr)
eblcase <- read.csv("C:/Users/HP PAVILLION 17/OneDrive/Desktop/Mult/eblcat (3).csv")
str (eblcase)
eblcase$black <- ifelse(eblcase$racecat == 1,1,0)
eblcase$asian <- ifelse(eblcase$racecat == 2,1,0)
```

b) Write down the population model(s) that correspond with the plan you propose in part (a). (7 points)
   - $\boldsymbol{\mu_{EBL}} = \boldsymbol{\beta_0} + \boldsymbol{\beta_1}(\mathbf{les_{size}}) + \boldsymbol{\beta_2}(\text{black}) + \boldsymbol{\beta_3}(\text{asian}) + \boldsymbol{\beta_4}(\text{hispanic}) + \boldsymbol{\beta_5}(\mathbf{les_{size}}) * (\text{black}) + \boldsymbol{\beta_6}(\mathbf{les_{size}}) * (\text{asian}) + \boldsymbol{\beta_7}(\mathbf{les_{size}}) * (\text{hispanic})$
   -
```
modelrace2 <- lm(ebl ~ les_size + black + asian + hispanic +les_size:black + les_size:asian + les_size:hispanic , data = eblcase)
summary (modelrace2)
```

c) Follow the plan you outlined in part (b). In addition to including the appropriate output, write down the estimated model(s) you fit. (7 points)
   - $\boldsymbol{\mu_{EBL}} = 588.79^{***} + 20.29(\mathbf{les_{size}})^{***} - 5.641(\text{black}) + 144.0(\text{asian}) + 31.69(\text{hispanic}) - 0.144(\mathbf{les_{size}}) * (\text{black}) - 71.9(\mathbf{les_{size}}) * (\text{asian}) - 11.861(\mathbf{les_{size}}) * (\text{hispanic})$
   -
```
lm(formula = ebl ~ les_size + black + asian + hispanic + les_size:black +
    les_size:asian + les_size:hispanic, data = eblcase)

Residuals:
     Min       1Q   Median       3Q      Max
-270.822  -48.942    4.579   56.461  231.280

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       588.792     19.518  30.167  < 2e-16 ***
les_size           20.290      5.499   3.690 0.000326 ***
black              -5.641     61.006  -0.092 0.926462
asian             144.020    110.904   1.299 0.196310
hispanic           31.594     40.188   0.786 0.433159
les_size:black     -0.144     10.922  -0.013 0.989497
les_size:asian    -71.910     45.632  -1.576 0.117411
les_size:hispanic -11.861      8.699  -1.363 0.175012
```
   -
   -

d) Test whether the model for estimating blood loss based on an interaction model with tumor size and race is meaningful on level of significance 5%. State the appropriate null and alternative hypothesis, test statistics and associated dfs, p-value, conclusion, and interpretation. Include the appropriate output in the answer. (7 points)

```
Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)        588.792     19.518  30.167  < 2e-16 ***
les_size            20.290      5.499   3.690 0.000326 ***
black               -5.641     61.006  -0.092 0.926462
asian              144.020    110.904   1.299 0.196310
hispanic            31.594     40.188   0.786 0.433159
les_size:black      -0.144     10.922  -0.013 0.989497
les_size:asian     -71.910     45.632  -1.576 0.117411
les_size:hispanic  -11.861      8.699  -1.363 0.175012
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 95.83 on 134 degrees of freedom
Multiple R-squared:  0.145,     Adjusted R-squared:  0.1004
F-statistic: 3.247 on 7 and 134 DF,  p-value: 0.003248
```

- **Omnibus F-test:**

- **H_0: all betas = 0 (The model is not meaningful) H_A: not H_0 (The model is meaningful)**
- **$F_{obs} = 3.247 \sim$ F-distribution with $df_1 = 7$ and $df_2 = 134$**

- **Since p-value = $P(F_{7,134} > 3.247) = 0.003248 < 0.05 = \alpha$, we reject the H_0.**

- **At the 5% level of significance, the data provide enough evidence to conclude that the model is meaningful ($F_{obs} = 3.247$, $df_1 = 7$ and $df_2 = 134$, p = 0.003248).**

-value Conclusion: The model is significant for this. As p value for omnibus test significant i.e. is p= less than 5%

Interpretation: It means at least one beta/coefficient is significant among all coefficients.

e) Test whether the interaction effect is significant. State the appropriate null and alternative hypotheses, test statistic and associated dfs, p-value, conclusion, and interpretation. Include the appropriate output in the answer. (7 points)

We'll do a partial likelihood test for this question to check the appropriate whether interaction is significant or not. For this we'll run an OLS regression model without interaction and one with interaction (which we did in part b) and later run a likelihood ratio test with two models

```
modelrace1 <- lm(ebl ~ les_size + black + asian + hispanic, data = eblcase)
modelrace2 <- lm(ebl ~ les_size + black + asian + hispanic +les_size:black + les_size:asian + les_size:hispanic , data = eblcase)
summary (modelrace2)
summary (modelrace1)
lmtest::lrtest(modelrace1, modelrace2)
```

```
> lmtest::lrtest(modelrace1, modelrace2)
Likelihood ratio test

Model 1: ebl ~ les_size + black + asian + hispanic
Model 2: ebl ~ les_size + black + asian + hispanic + les_size:black +
    les_size:asian + les_size:hispanic
  #Df  LogLik Df  Chisq Pr(>Chisq)
1   6 -847.49
2   9 -845.25  3 4.4746     0.2146
> |
```

- Reduced Model
- $\boldsymbol{\mu_{EBL} = \beta_0 + \beta_1(\text{les}_{\text{size}}) + \beta_2(\text{black}) + \beta_3(\text{asian}) + \beta_4(\text{hispanic})}$
- $\boldsymbol{\mu_{EBL}} = 602.678^{***} + 15.8(\text{les}_{\text{size}})^{***} + 1.878(\text{black}) - 21.635(\text{asian}) - 14.03(\text{hispanic})$

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  602.678     15.328  39.320  < 2e-16 ***
les_size      15.832      3.887   4.073 7.82e-05 ***
black          1.878     36.035   0.052    0.959
asian        -21.635     40.552  -0.534    0.595
hispanic     -14.029     22.628  -0.620    0.536
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 96.28 on 137 degrees of freedom
Multiple R-squared:  0.1177,    Adjusted R-squared:  0.09189
F-statistic: 4.567 on 4 and 137 DF,  p-value: 0.001712
```

- Reduce model without interaction term is also significant with Omnibus test.
- **F_{obs} = 4.567 ~ F-distribution with df_1 = 4 and df_2 = 137**
- **Since p-value = $P(F_{4,137} > 3.247)$ = 0.001712 < 0.05 = α, we reject the H_0.**
- **At the 5% level of significance, the data provide enough evidence to conclude that the model is meaningful (F_{obs} = 4.567, df_1 = 4 and df_2 = 137, p = 0.001712).**
- A partial likelihood test would tell us whether we must use interaction of race and size at all.

- Partial Likelihood Ratio Test

- H_0: $\beta_5 = \beta_6 = \beta_7 = 0$, i.e., reduced model is sufficient.
- H_A: $at\ least\ one\ slope\ in\ (\beta_5, \beta_6, \beta_7)\ is\ non-zero$ Full model is better than reduced model.
- Test statistic, $\chi^2_{df=3} = 4.4746$
- With $df = 3,\ p\ value = 0.2146 > 0.05$ → FAIL TO REJECT H_0

- The data does not provide enough support to conclude that the full model is better than the reduced model. Therefore, the interaction between tumor size and race on outcome blood loss is not meaningful at the 5% level of significance.
-

f) Summarize the results of your modeling approach in parts (b) - (e) in a short paragraph (no more than 10-12 sentences) (7 points)

Here is some guidance that you can use: Describe the variables used for the analysis, then talk about what statistical tool(s) and tests you used and for what purpose. The results and conclusions from the analysis need to be presented in line with the purpose of the research. More generally, you could talk about whether each model is a good fit to the data, i.e., meaningful, and what proportion of variance of the outcome is explained by what predictors. Next, you could talk about each predictor, whether they are significant and if yes, you would interpret them.

- **Part b**
- $\boldsymbol{\mu}_{\boldsymbol{EBL}} = 588.79^{***} + 20.29(\mathbf{les}_{\mathbf{size}})^{***}$ -5.641(black) + 144.0(asian) + 31.69(hispanic) - 0.144$(\mathbf{les}_{\mathbf{size}}) *$ (black) -71.9$(\mathbf{les}_{\mathbf{size}}) *$ (asian) -11.861$(\mathbf{les}_{\mathbf{size}}) *$ (hispanic)
- In part b we used an interaction term to find the impact of race with tumor size on blood loss.
- We first generated dummy variables for race with k-1, 3 dummies for four categories of race/ethnicity. After running the OLS regression, the Omnibus test gives us p value 0.003 which means model containing race, size and interaction of all races with tumor size is meaningful. Moreover, the multiple R-squared only tells us 0.14% for the above model and does not explain for the more that 99% of blood loss. However, after looking at the p value of all variables and their interactions only size is significant on the individual level see below the output as well. But this does not give us the enough evidence to drop race from our model.
- Reduced model takes only race with size without interaction and this model with omnibus test is also significant with p-value 0.0017.
- Likelihood, test compares the two models and tell us whether interaction model is better or not and chi-square value of 0.2146 which is greater than 5% likelihood test us that reduced model is better. We do not need race interaction at all.
- What about race?
- Now, we can also see the from OLS results of reduce models that although Omnibus test tells us that model is significant, but none of the race dummy variables is significant which means race is not important for blood loss with the tumor size variables.
- It means blood loss in ml would increase by 20.29ml with increase of 1 cm in tumor size .
- None of the race or race interaction is significant so race does not matter for this model.
- **Part e**
- Model e test the validity of interaction of race and size with only race and results suggest we do not have to use interaction of race and size.

```
> lmtest::lrtest(modelrace1, modelrace2)
Likelihood ratio test

Model 1: ebl ~ les_size + black + asian + hispanic
Model 2: ebl ~ les_size + black + asian + hispanic + les_size:black +
    les_size:asian + les_size:hispanic
  #Df  LogLik Df  Chisq Pr(>Chisq)
1   6 -847.49
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```

```
setwd("C:/Users/HP PAVILLION 17/OneDrive/Desktop/Mult")
library(readr)
eblcase <- read.csv("C:/Users/HP PAVILLION 17/OneDrive/Desktop/Mult/eblcat (3).csv")
str (eblcase)
eblcase$black <- ifelse(eblcase$racecat == 1,1,0)
eblcase$asian <- ifelse(eblcase$racecat == 2,1,0)
eblcase$hispanic <- ifelse(eblcase$racecat == 3,1,0)
install.packages("lmtest")
install.packages("car")
install.packages("MASS")
install.packages("lme4")
install.packages("glmnet")
install.packages("lmtest")
library(lmtest)
str(eblcase)
modelrace1 <- lm(ebl ~ les_size + black + asian + hispanic, data = eblcase)
modelrace2 <- lm(ebl ~ les_size + black + asian + hispanic +les_size:black + les_size:asian + les_size:hispanic , data =
summary (modelrace2)
summary (modelrace1)
lmtest::lrtest(modelrace1, modelrace2)
```